

Boss of the SOC Version 1 (2015)

10 May 2026 23:30

Ransomware

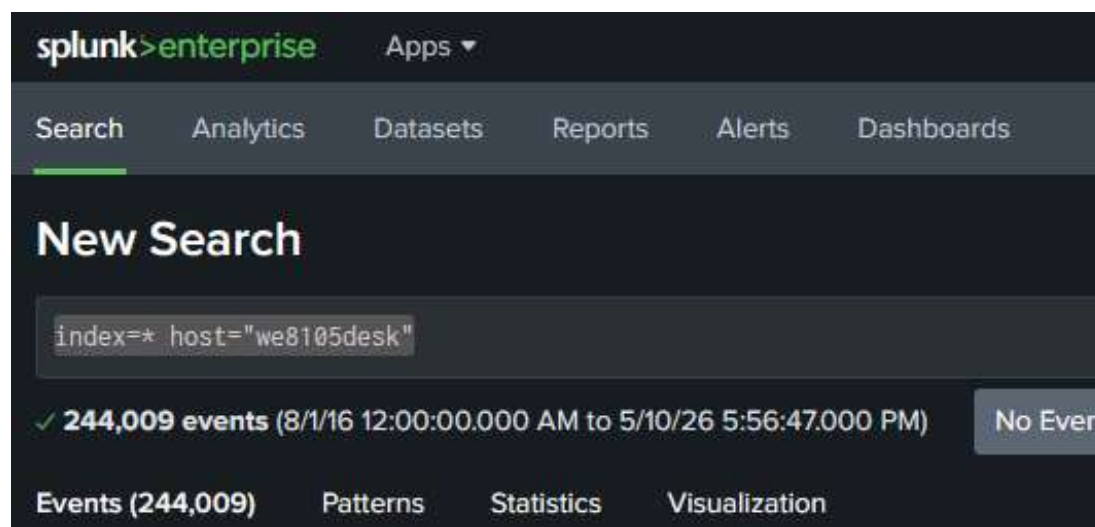
After the excitement of yesterday, Alice has started to settle into her new job. Sadly, she realizes her new colleagues may not be the crack cybersecurity team that she was led to believe before she joined. Looking through her incident ticketing queue she notices a "critical" ticket that was never addressed. Shaking her head, she begins to investigate. Apparently on August 24th Bob Smith (using a Windows 10 workstation named we8105desk) came back to his desk after working-out and found his speakers blaring (click below to listen), his desktop image changed (see below) and his files inaccessible.

Alice has seen this before... ransomware. After a quick conversation with Bob, Alice determines that Bob found a USB drive in the parking lot earlier in the day, plugged it into his desktop, and opened up a word document on the USB drive called "Miranda_Tate_unveiled.dotm". With a resigned sigh she begins to dig into the problem...

#200

Question:

What was the most likely IPv4 address of we8105desk on 24AUG2016?

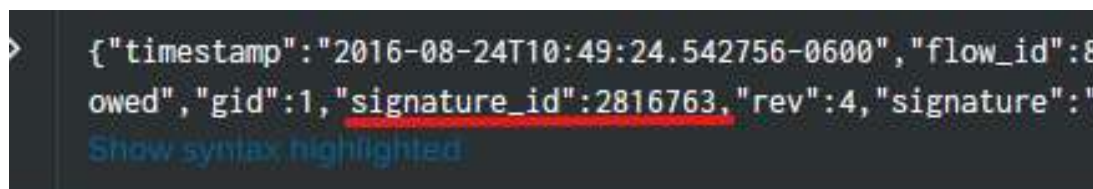
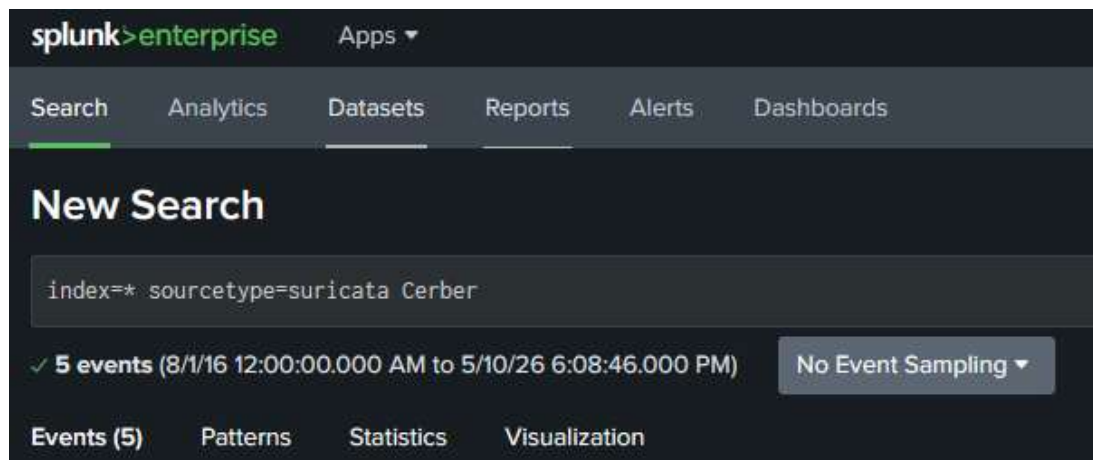


Top 10 Values	Count	%
192.168.250.100	53,106	97.34%
192.168.2.50	1,217	2.231%
127.0.0.1	105	0.192%
192.168.250.255	70	0.128%
0.0.0.0	42	0.077%
224.0.0.252	6	0.011%
c0a8:fa64:0:0:9820:26fc:1e0:ffff	4	0.007%
::1	3	0.005%
fe80:0:0:0:9dac:222e:c1f8:d3d8	3	0.005%
239.255.255.250	1	0.002%

#201

Question:

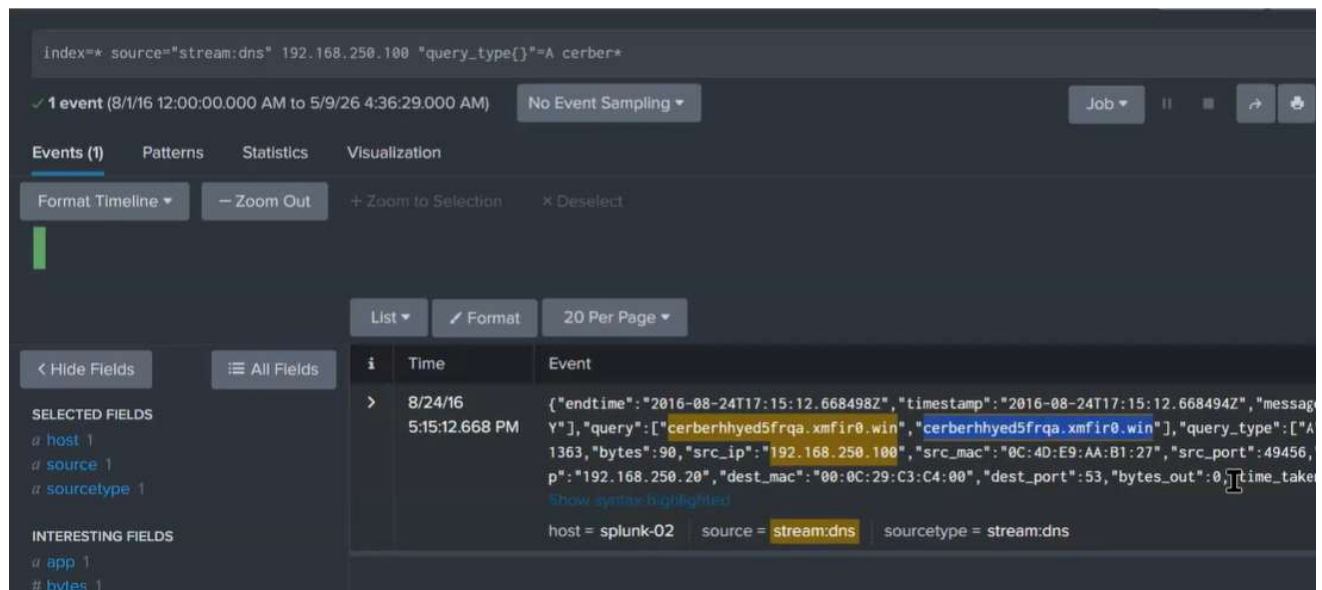
Amongst the Suricata signatures that detected the Cerber malware, which one alerted the fewest number of times? Submit ONLY the signature ID value as the answer.



#202

Question:

What fully qualified domain name (FQDN) does the Cerber ransomware attempt to direct the user to at the end of its encryption phase?



#203

Question:

What was the first suspicious domain visited by we8105desk on 24AUG2016?

splunk>enterprise Apps ▾

Search Analytics Datasets Reports Alerts Dashboards

New Search

```
index=* source="stream:dns" "query_type{}"="A "192.168.250.100"
NOT query()=*.microsoft.com
NOT query()=*.msn.com
NOT query()=*.windows.com
NOT query()=*.office.com
NOT query()=*.live.com
NOT query()=*.outlook.com
NOT query()=*.bing.com
NOT query()=*.skype.com
NOT query()=*.zoom.com
| table _time, query{}, query_type{}, dest_ip
| sort _time
```

✓ 131 events (8/1/16 12:00:00.000 AM to 7/1/26 10:29:21.000 AM) No Event Sampling ▾

#204

Question:

During the initial Cerber infection a VB script is run. The entire script from this execution, pre-pended by the name of the launching .exe, can be found in a field in Splunk. What is the length of the value of this field?

New Search Save As

```
index=* sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" *.vbs app="C:\\windows\\syswow64\\cmd.exe"
| eval lencmd=len(CommandLine)
| table lencmd
```

✓ 2 events (8/1/16 12:00:00.000 AM to 7/1/26 10:38:28.000 AM) No Event Sampling ▾ Job ▾ || ■

Events Patterns **Statistics (2)** Visualization

100 Per Page ▾ Format Preview ▾

lencmd ↕

102
4490

#205

Question:

What is the name of the USB key inserted by Bob Smith?

New Search

```
index=* sourcetype="winregistry" host=we8105desk *usb*
| table registry_value_data
| dedup registry_value_data
```

✓ 519 events (8/1/16 12:00:00.000 AM to 7/1/26 10:43:26.000 AM) No Event Sampling ▾

Events Patterns **Statistics (30)** Visualization

100 Per Page ▾ Format Preview ▾

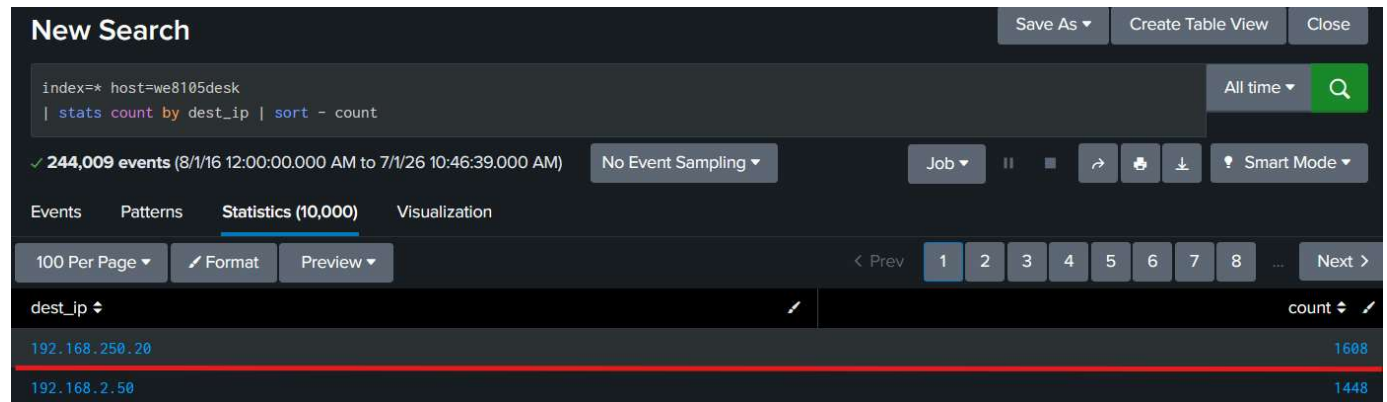
registry_value_data ↕

LowDateTime:803713417,HighDateTime:0***Binary mof compiled successfully
0x000000a4(164)
0x00000001(1)
MIRANDA_PRI
\\?\\WpdBusEnumRoot#UMB#2837c186b08&STORAGE#VOLUME#_??_USBSTOR#DISK&VEN_GENERIC&PROD_FLASH_DISK&REV_8.07#7D961196&0#{f33fdc04-d1ac-4e8e-9a30-19bdd4b108ae}

#206

Question:

Bob Smith's workstation (we8105desk) was connected to a file server during the ransomware outbreak. What is the IPv4 address of the file server?



The screenshot shows the Splunk Search interface with the following details:

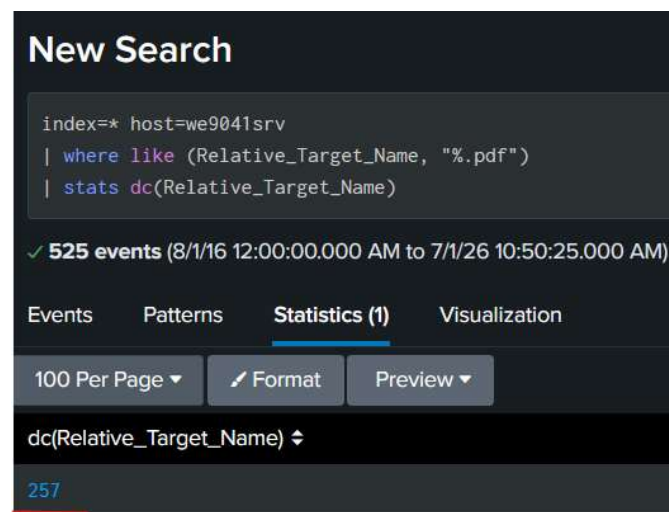
- Search Bar:** `index=* host=we8105desk`
`| stats count by dest_ip | sort - count`
- Results:** 244,009 events (8/1/16 12:00:00.000 AM to 7/1/26 10:46:39.000 AM)
- Table View:** Columns are `dest_ip` and `count`.

dest_ip	count
192.168.250.20	1688
192.168.2.50	1448

#207

Question:

How many distinct PDFs did the ransomware encrypt on the remote file server?



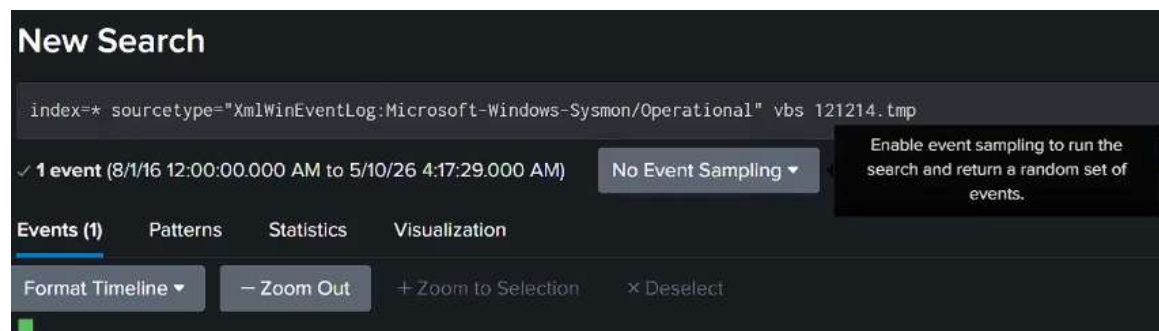
The screenshot shows the Splunk Search interface with the following details:

- Search Bar:** `index=* host=we9041srv`
`| where like (Relative_Target_Name, "%.pdf")`
`| stats dc(Relative_Target_Name)`
- Results:** 525 events (8/1/16 12:00:00.000 AM to 7/1/26 10:50:25.000 AM)
- Table View:** Column is `dc(Relative_Target_Name)` with a value of 257.

#208

Question:

The VBScript found in question 204 launches 121214.tmp. What is the ParentProcessId of this initial launch?



The screenshot shows the Splunk Search interface with the following details:

- Search Bar:** `index=* sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" vbs 121214.tmp`
- Results:** 1 event (8/1/16 12:00:00.000 AM to 5/10/26 4:17:29.000 AM)
- Table View:** Shows a single event in the timeline view.

ParentCommandLine	"C:\Windows\System32\WScript.exe" "C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\20429.vbs"
ParentImage	C:\Windows\SysWOW64\wscript.exe
ParentProcessGuid	{0F2D76F0-CEA9-57BD-0000-0010C8133100}
ParentProcessId	3968
ProcessGuid	{0F2D76F0-CFD5-57BD-0000-0010E3AC3400}
ProcessId	1476
RecordId	364394
SHA1	EE8CBF12D87C4D388F09B4F69BED2E91682920B5
SHA256	17F746D82695FA9B35493B41859D39D786D32B23A9D2E00F4011DEC7A02402AE
SecurityID	S-1-5-18
Task	1

#209

Question:

The Cerber ransomware encrypts files located in Bob Smith's Windows profile. How many .txt files does it encrypt?

New Search

```
index=* sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"
TargetFilename="C:\\Users\\bob.smith.WAYNECORPINC\\*.txt"
| stats dc(TargetFilename) AS distinct_txt_files_encrypted
```

! The term 'TargetFilename="C:\\Users\\bob.smith.WAYNECORPINC*.txt"' contains a wildcard. Wildcard representations include punctuation. [Learn More](#)

✓ **406 events** (8/1/16 12:00:00.000 AM to 5/10/26 4:47:27.000 AM) No Event Sampling

Events Patterns **Statistics (1)** Visualization

20 Per Page Format Preview

distinct_txt_files_encrypted

406

#210

Question:

The malware downloads a file that contains the Cerber ransomware cryptor code. What is the name of that file?

New Search

Save As Create Table View Close

index=* sourcetype="suricata" dest_ip="192.168.250.100" file_state=CLOSED "http.hostname"="*.org"

All time

Q

1 event (8/1/16 12:00:00.000 AM to 7/1/26 10:57:15.000 AM) No Event Sampling

Job

||

Smart Mode

Events (1) Patterns Statistics Visualization

Format Timeline Zoom Out + Zoom to Selection X Deselect

1 month per column

List Format 20 Per Page

Hide Fields All Fields

SELECTED FIELDS

a host 1

a source 1

a sourcetype 1

INTERESTING FIELDS

a app 1

i

Time

Event

>

8/24/16 4:48:21.634 PM

{"timestamp": "2016-08-24T10:48:21.634085-0600", "flow_id": 4129315976, "event_type": "fileinfo", "src_ip": "37.187.37.150", "src_port": 80, "dest_ip": "192.168.250.100", "dest_port": 49407, "proto": "TCP", "http": {"hostname": "solidaritedeproximate.org", "url": "\mhtr.jpg", "http_user_agent": "Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/4.0; SLCC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC 6.0; InfoPath.3)", "http_content_type": "text/html", "http_method": "GET", "protocol": "HTTP/1.1", "status": 404, "length": 1760, "app_proto": "http", "fileinfo": {"filename": "\mhtr.jpg", "s"
Show syntax highlighted

host = suricata-ids.waynecorpinc.local

source = /var/log/suricata/eve.json

sourcetype = suricata

#211

Question:

Now that you know the name of the ransomware's encryptor file, what obfuscation technique does it likely use?
Steganography